PrahaarShield – Virtual Jury Prep

Nov 4, 2026 · @Anushka

Today first: find your slot

Assume the jury call could be today (4 Nov): be ready from 9:00 on both days until the slot mail arrives.

1. **Search Gmail, every folder** (Spam, Promotions, Updates, All Mail) for: `SFIC`, `Seva`, `jury`, `slot`, `meeting`, `Meet`, `Zoom`, `Teams`. Check the exact email address you registered with on the portal.
2. **Portal:** app.sevainnovationchallenge.in → participant dashboard → Activity and any notifications.
3. **Phone:** SMS and WhatsApp on your registered number; keep the ringer on.
4. **Reply in the same thread** as the shortlist email (template below). Do not start a new email; a reply in the thread is found faster.
5. **No answer by 11:00:** use the phone number or email on the website's Contact / FAQ page.
6. **Both days:** keep 9:00–18:00 free, laptop charged and the app already running, dressed formally from the morning.

Reply template:

> Subject: Re: Application Shortlisted – Virtual Jury Evaluation – PRAHAARSHIELD
>
> Dear SFIC Organising Team,
>
> Thank you for shortlisting my application PRAHAARSHIELD (Sector 9, Responsible AI & Digital Trust). I have not yet received my jury date, time slot or meeting link, and I wanted to make sure I do not miss it. Could you please share my schedule?
>
> Registered name: Anushka Anand Bhakare · Registered email: [your email] · Phone: [your number]
>
> I am available on both 4th and 5th November.
>
> Thank you, Anushka Bhakare

It may be on either day. If the link arrives late, join immediately; a short delay is understandable, missing the call is not.

What the jury scores, and your proof

Reviewers score innovation, relevance, technical soundness and impact (startupgrantsindia summary); the official guide adds a judges' lens: real evidence, a working demo, a fit with money the government already spends, and a deliverable pilot plan.

Criterion	What they check	Your proof (say it in one line)
Relevance	Is the problem real, big, local?	₹91,000 lost per complaint (2025); Maharashtra lost ₹3,203 cr in 2025, the highest of any state
Innovation	What is new versus what exists?	Reads the whole chat; knows relationships; covers saved contacts (WhatsApp Scam Alert checks only non-contacts); explains in Marathi and Hindi
Technical soundness	Does it work, and was it measured honestly?	Live demo; 94.4% of scams caught, 3.3% false alarms on a held-out test used once; several seeds; failures shown
Impact & feasibility	Can it be delivered, by whom, at what cost?	20 → 200 → 5,000 users → state; owner: Maharashtra Cyber; pilot ≈ ₹2.4 lakh; ₹0 cloud cost per user
Judges' lens: one failure	Can you say what failed?	Our-data-only model scored 0.75 on another writer's messages → added outside data → 0.93
Judges' lens: govt money	Does it reduce an existing cost?	Prevents losses before payment, so fewer complaints, 1930 calls and recovery cases for cyber police

The safest stance throughout: confident about what is built and measured, precise about what is planned, honest about limits.

Competition structure and your women category

The number of applicants shortlisted, in total or by category, has not been published anywhere I could find, and your email gives none; do not rely on any figure for it.

What is published (Free Press Journal, official site, startupgrantsindia):

Item	Published fact
Registrations, Maharashtra zone	65,000+ registrations from 48,680 teams
Women participants / under 18	3,787 women; 2,044 under 18
General category	12,604 reported as general-category registrations (this figure does not add up with the others in the report, so treat it as unclear)
Winners per sector	5: three General, one Women, one Below-18; ₹10 lakh work order each
How the Women category works	You choose General or Women at profile stage; Women entries are ranked separately within each sector
Finalists / awardees	about 200 exhibition finalists; 50 awardees across 10 sectors
Process	administrative scrutiny, two independent online evaluations, technical and impact assessment, finalist validation, Grand Jury
Final event	Nagpur, 7 December 2026; shortlisted participants present and demonstrate before the jury (FPJ)

What it means for you: in Sector 9 (Emerging Technologies) your direct competition for the women slot is the other women-category entries in that sector, and the jury still scores everyone on the same criteria. Being shortlisted already puts you past administrative scrutiny and the two online evaluations. Because the final stage asks for a demonstration, the live demo matters most.

WhatsApp Scam Alert: what Meta declared, the gap, and our answer

Meta's engineering post (12 Aug 2026) describes an optional, on-device feature in limited beta; the table compares it with what we do and plan. Be fair to Meta: it is strong on privacy engineering and scale, and our claim is about coverage and explanation, not about a better model.

Area	What Meta declared	The gap	PrahaarShield today	PrahaarShield planned
Who is checked	Incoming messages from non-contacts	Scams from saved contacts (hijacked accounts, a fake boss on a known number) are outside the description	Checks saved contacts too, with relationship-aware rules and a "call them on the number you already have" step	Same on the phone; user can set how much a contact is trusted
How it decides	On-device ML on conversational structure and linguistic signals; trained on scam conversations users reported	Languages and countries covered are not stated	MuRIL fine-tuned on English, Hindi, Marathi and Hinglish; six Indian financial scam types; tactics and facts remembered across the chat	Real-message data from the pilot, with consent; more Indian languages

Area	What Meta declared	The gap	PrahaarShield today	PrahaarShield planned
What the user sees	A warning in the chat, visible only to the user; options to block, report or continue; mark chat as trusted	The description gives no reasons for the warning; a chat marked trusted is never flagged again, so a hijacked chat can go quiet	Five levels; red card with reasons; SOS in English, Hindi, Marathi; fact checks; "Not a scam" quiets alerts, but hard rules (OTP, "safe account", WhatsApp code) still fire	User study on whether people understand and act on the explanations
Privacy	No message content leaves the device for classification; no automatic reporting; activity log under Request Info; model weights published; model versions on a public ledger; only aggregated counts leave the device	None: this is the standard to match	Analysis is local in our design; code is public on GitHub	Model card, published model, in-app activity log
Reporting	User can report to WhatsApp	No link to Indian authorities is described	1930, cybercrime.gov.in, Chakshu (Sanchar Saathi), evidence pack	Consent-based hand-off to Maharashtra Cyber / I4C
What is checked	Messages	Images, QR codes and links are not mentioned	Text, plus link, UPI and number checks	OCR for screenshots and notices, UPI QR decoding, link reputation

Area	What Meta declared	The gap	PrahaarShield today	PrahaarShield planned
Status	Limited beta	Not yet available to all users	Lab prototype and simulator	Android phone pilot, then integration through partnership

Where Meta is stronger (say this before the jury does): native integration inside WhatsApp, scale, a large set of real user-reported scams for training, published weights, an independent transparency ledger and a bug-bounty programme.

Our one-line positioning: "Meta is building a general scam alert; we are building the Indian-language, saved-contact-aware, explain-and-act layer that connects to 1930 and Maharashtra Cyber."

Pitch scripts

Lead with the problem, show the demo early, and end with the ask; speak slowly, about 130 words a minute.

60-second version (memorise this):

> Good morning. I'm Anushka Bhakare, a third-year Computer Engineering student at Cummins College, Pune. Every year Indians lose thousands of crores to WhatsApp scams; in 2025 the average loss was about ₹91,000 per complaint, and Maharashtra lost the most of any state. These scams don't come in one message. A fake officer builds trust, then pressure, then demands money, in Hindi, Marathi or Hinglish. PrahaarShield is a scam-guard assistant designed to live inside WhatsApp. It reads the whole conversation on the phone, understands who the sender is to you, checks claims against verified facts, and warns you before you pay, in your own language. Our model catches 94.4% of scams in testing with 3.3% false alarms. I'd like to show it working, and then our plan to pilot it with Pune City Cyber Police.

5-minute version, slide by slide:

1. **Title (15 s):** name, college, one line: "a scam guard that lives inside WhatsApp and warns you before you pay".

2. **Problem (40 s):** read the three scam messages aloud (approach, pressure, demand). "Each looks harmless alone." Then ₹91,000, ₹3,203 crore, and "only about 3% become FIRs, so money rarely comes back."
3. **Why it keeps working (30 s):** the five whys end at "no one checks the whole chat, in your language, at the moment of payment". Posters are forgotten, blocklists miss hijacked contacts, cloud AI breaks privacy, and WhatsApp's own Scam Alert checks only non-contacts.
4. **The assistant (40 s):** ⋮ menu (scan this chat, all chats, new chats), live checks, SOS button, five levels. "Quiet until it matters." Privacy: analysis on the phone, nothing leaves the device. Say clearly: "this is the proposed integration; today it runs in our WhatsApp-style simulator".
5. **How it works (30 s):** seven stages in one sentence each; stress "context like contact status goes to the Brain, never to the model".
6. **Inside the AI (30 s):** MuRIL from Google, fine-tuned to answer four questions at once; the six pillars are why it can explain itself.
7. **The Brain (30 s):** read the Raj example output: it caught a hacked-contact scam even though every single message looked normal.
8. **How we built it (30 s):** "every choice was tested": one failure (0.75 → 0.93), what we dropped and why.
9. **Results + demo (60 s):** switch to the live demo here (next section). Then the numbers, and the honest limits line.
10. **Roadmap and ask (35 s):** phone app, images and QR next; pilot path 20 → 200 → 5,000 → state; owner Maharashtra Cyber; ask: mentorship, a pilot, funding for the Android build and a real-message test.

Closing line: "PrahaarShield is built, measured and ready for a real-world pilot. With Maharashtra Cyber, we can stop the loss before the payment, not after. Thank you."

If they give only 3 minutes: slides 2, 4, a 60-second demo, 9, 10.

Live demo plan (3–4 minutes)

Three short chats prove three different things: it catches a classic scam, it catches a hijacked friend no keyword filter would, and it stays quiet in a normal family chat. Rehearse each once on the morning of the call; keep this page open on your phone.

Before the call: app running, `http://localhost:8000/sim` open in one browser tab (both sides visible), zoom 110–125%, every other tab closed. Between demos press the reset arrow; change setup with the sliders icon.

Demo 1: Digital arrest (unknown number, English explanations)

Setup: General · contacts Yes · Not in contacts · `+91 70XXX 45210`. Type on the scammer side:

1. `Main Mumbai Cyber Crime Branch se Inspector Rathod bol raha hu.`
2. `Aapke Aadhaar se money laundering case darj hua hai.`
3. `Ye confidential hai. Kisi ko mat batana aur video call pe bane rehna, warna aaj hi arrest hoga.`
4. `Innocent prove karne ke liye RBI safe account me abhi ₹2,80,000 transfer karo.`

Say: "Notice it stays quiet at first, warns at the threat, and at the demand shows a full-screen STOP." Then press SOS and point at: where you are in the scam, red flags, the fact check "RBI has no safe account", Call 1930.

Demo 2: Hijacked saved contact (Marathi explanations)

Setup: language मराठी · General · contacts Yes · **Safe (saved contact)** · name `Raj`. Type on the scammer side, with your replies on the user side:

1. Raj: `Lily मला 13000 पाठव पटकन`
2. You: `Ho pan ka have aahet`
3. Raj: `अग पटकन मला urgent loan रिपेमेंट साठी हवे आहेत 35000 होते, 5 मिनिटात नाही पाठवले तर अकाउंट freeze करतील`
4. You: `Thamb mi tula call karte`
5. Raj: `अरे आत्ता call नको करू. मी bank च्या customer care शी बोलतोय`
6. Raj: `He बघ पैसे पाठव नाहीतर परिणाम वाईट होतील`
7. Raj: `otp send kar`

Say: "Raj is a saved contact, and each message looks normal, which is exactly how hijacked accounts work. The Brain adds it up: growing amounts, a blocked call, a threat, then an OTP request." Press SOS: the reasons appear in Marathi, including "call Raj on the number you already have".

Demo 3: Normal family chat stays safe

Setup: contacts Yes · Safe (saved contact) · name `Aai`. Type: `बाळा, उद्या कॉलेज फीची शेवटची तारीख आहे, नाहीतर ₹500 दंड लागेल` then `Amazon चा delivery boy आलाय, OTP आला असेल तो सांग पटकन`.

Say: "A fine, money and even an OTP, and no alarm, because it understands a delivery OTP at home is normal. A bank or WhatsApp code would be flagged even from family."

Optional extras if time allows: switch the EN tag to हिंदी and मराठी to show all three languages; type `sorry yaar galti se tujhe 6 digit code bheja, forward kar de` from a saved friend to show the WhatsApp-hijack rule.

If something fails:

- App frozen or slow: say "let me show you the recorded run", then use slide 9 screenshots (or your demo video).
- A message is judged differently from rehearsal: stay calm and use it. "This is a real model, not a script; this is exactly the kind of case our field pilot will measure."
- Internet drops: rejoin from phone hotspot; keep the PDF deck open as a fallback.
- Screen share fails: share the browser tab instead of the whole screen, or present the PDF.

Tech and call setup checklist

Start the app 20 minutes before your slot and confirm it says threshold 0.86; everything else is preparation you can finish tonight.

The app (20 min before):

```
cd C:\PRahaarshield_Miniproject\app
$env:PRAHAAR_MODEL_DIR="C:\PRahaarshield_Miniproject\models_final"
.\run_app.bat
```

- [ ] Terminal shows `model ready (threshold 0.86…)`
- [ ] http://localhost:8000/api/info opens and shows the engine
- [ ] Demo 1 sent once as a warm-up (the first message after start is the slowest), then reset
- [ ] Terminal minimised, not closed

Laptop:

- [ ] Charger plugged in, battery saver off

- [ ] Windows Focus / Do Not Disturb on; WhatsApp Desktop, mail and chat apps closed
- [ ] Only these open: the meeting, the demo tab, the deck PDF, this prep doc
- [ ] Desktop and browser bookmarks bar free of personal or unrelated items
- [ ] Windows updates paused for the day

Internet and call:

- [ ] Wi-Fi tested; phone hotspot ready as backup
- [ ] Meeting app (Meet, Zoom or Teams) installed and tested; in Chrome, screen-share permission allowed
- [ ] Display name: `Anushka Bhakare – PRAHAARSHIELD`
- [ ] Practise sharing a single browser tab or window, and switching between the deck PDF and the demo tab
- [ ] Join 10 minutes early; camera on, mic muted until spoken to

You and the room:

- [ ] Earphones with mic; camera at eye level; light in front of you, not behind
- [ ] Quiet room, door closed; family informed of the slot time
- [ ] Formal clothes; water nearby; college ID and Aadhaar within reach in case they verify identity
- [ ] Notebook and pen to write down each question before answering

Numbers cheat sheet

Know these cold; if asked for a source, name it, and if you are unsure of a number, say "about" rather than guess.

Topic	Number	Where it comes from
Loss per complaint, 2025	≈ ₹91,000	I4C 2025 data (IANS): ₹19,813 cr ÷ 21.8 lakh complaints
Maharashtra loss, 2025	₹3,203 cr (highest state)	I4C 2025 data (IANS)
Complaints 2021–25	65.89 lakh; ₹55,050 cr reported lost	I4C / NCRP, Lok Sabha written reply
FIRs	≈ 3% (1,95,760 of 65.89 lakh)	same Lok Sabha reply

Topic	Number	Where it comes from
Money saved by 1930 / CFCFRMS	₹11,158 cr (to June 2026)	same Lok Sabha reply
WhatsApp Scam Alert	limited beta, 12 Aug 2026; non-contacts only; on-device	Meta engineering blog
Scams caught (recall)	94.4% (95% range 90.6–97.6%)	our held-out test, 434 messages, 160 scams
False alarms	3.3%	same test, 274 normal messages
Precision / scam F1 / macro-F1	94.4% / 0.944 / 0.956	same test
Calibration error (ECE)	0.023	same test
By language (macro-F1)	EN 0.93 · HI 0.98 · Hinglish 0.98 · MR 0.95	same test
Scam-type accuracy (macro-F1)	0.92	same test
Threshold used by the app	0.86 (chosen on validation)	validation: ≥ 95% scams caught, 2.6% false alarms
Outside-writer test	0.75 (our data only) → 0.93 (with outside data)	our experiment, 3 seeds
Self-test (63 messages, held out)	33/34 scams caught; 1/29 false alarms	`selftest.py`
Test chats (20, hand-written)	14/14 scam chats Dangerous; 12/14 Critical; 0/6 normal alerted	`selftest.py`
Brain relationship cases	10/10 correct with model scores forced low	Brain unit tests
Link / UPI / phone checks	78/78 fakes flagged; 0/39 genuine flagged	developer-written cases
Data	4,288 labelled messages: 3,420 train / 434 validation / 434 test	prepare_v3

Topic	Number	Where it comes from
Speed	≈ 65–150 ms per message on a laptop CPU	self-test median
Model	MuRIL base, ≈ 237 M parameters, ≈ 950 MB file	Google; our fine-tune
Pilot cost (estimate)	≈ ₹2.4 lakh; ₹0 cloud per user; ≈ ₹2,000 / month updates	our estimate
Break-even (illustrative)	1 prevented fraud ≈ 45 months of running cost	₹91,000 ÷ ₹2,000

Project deep dive

PrahaarShield is a learned model (MuRIL) inside a transparent pipeline: rules clean the text and check identifiers, the model reads intent, and a reasoning Brain decides over the whole chat.

The pipeline, per message:

1. **Normalizer:** Unicode NFKC; removes zero-width characters; maps look-alike letters (Cyrillic or Greek to Latin); joins s p a c e d letters; repairs defanged links (`hxxp` , `[.]`).
2. **Entity extraction:** finds URLs, UPI IDs, Indian and foreign phone numbers, amounts, account numbers, Aadhaar-like numbers and codes, then replaces them with placeholders (`<URL>` , `<AMOUNT>`) so the model learns the pattern, not a specific link or number.
3. **MuRIL multi-task model:** one encoder with four heads: scam / not scam, scam type (6), six intent pillars, language. Mean pooling over tokens, up to 256 tokens.
4. **Calibration:** temperature scaling per head, fitted on validation, so 0.9 really means about 90% likely; the decision threshold (0.86) is chosen on validation only.
5. **Hard rules:** an OTP or PIN request, an APK install, a UPI-PIN "collect" request, gift-card codes, a "safe account" or a "code sent by mistake" always reaches at least Dangerous; genuine warnings ("never share your OTP") are recognised and ignored.
6. **Identifier checks:** your contacts, the scam registry (simulated in the demo), official lists (TRAI 1600 government, 1601 financial institutions, 140 promotional series; .gov.in; bank UPI handles), look-alike domains (edit distance), and Indian number rules.

7. **Brain v3 (whole chat):** spots tactics (authority, urgency, threat, secrecy, familiarity, reward, links/apps); remembers money requests and escalation, blocked calls (only when repeated or while pushing payment), the kind of code asked for, and victim actions; applies relationship rules (a delivery OTP from family is normal; a bank or WhatsApp code from anyone is not); matches claims to 19 verified facts; outputs a level L0–L4 with reasons in English, Hindi or Marathi.

Training: fine-tuned on a free Colab T4 GPU with PyTorch and Hugging Face; AdamW (encoder lr 3e-5, heads 1e-3), 10% warmup, weight decay 0.01, mixed precision, batch 16, up to 8 epochs with early stopping; class weights and language weights; about 30 seconds per epoch.

Data (4,288 messages): our scam, normal and look-alike sets in 4 languages; 130 everyday money messages; 153 financial-threat scams plus 84 normal money threats; the public Ridham115 Indian scam SMS set (CC BY 4.0, outside writer). Lottery and courier scams are kept as "scam, no type". Split 80 / 10 / 10 by similarity groups so near-copies never cross from training into the test set.

Cleaning: removed 293 hidden interface tags that only appeared in normal messages (a label leak), reviewed 100 suspicious labels with an independent model (4 dropped), and removed training rows that resembled self-test cases.

How it was evaluated: options compared with several random seeds and judged only on validation; the winner tested once on the held-out test set, with 95% bootstrap ranges and results per language and per data source; then a separate hand-written self-test and 20 test conversations.

Why MuRIL: pre-trained by Google on 17 Indian languages including Roman-script text, which fits Hinglish and Marathi; strong published results on Indian-language benchmarks. HingRoBERTa-Mixed is being tested on Colab only as a comparison.

On a phone (planned): compress the model (distillation to a smaller student and 8-bit quantisation, typically about 4× smaller) and run it on an Arm processor; the Brain and rules are lightweight Python logic that ports easily.

Strict jury questions with answers

Answer in three beats: the direct answer, one piece of evidence, one honest limit or next step; under 40 seconds each.

Problem and evidence

Why this problem? I read The Hindu and Indian Express almost daily and see a scam story every one or two days, many on WhatsApp. The data agrees: about ₹91,000 lost per complaint in 2025, and Maharashtra lost the most of any state.

Did you talk to real victims or users? Not formally yet; my evidence today is official data and public advisories. Talking to users is the first step of the pilot: 20 volunteers in Pune who will also share, with consent, scam messages they receive.

Who exactly is your user? Any WhatsApp user in Maharashtra, but the design targets those hit hardest: elders, students, first-time digital users and people more comfortable in Hindi or Marathi.

Innovation and competition

WhatsApp already has Scam Alert. Why do we need you? It is a limited beta that checks only messages from non-contacts. Many scams come from hijacked contacts, like a boss or friend, which I4C warned about in 2026. Meta's model also looks at conversational structure, so our edge is different: we cover saved contacts, give reasons and fact checks in Hindi and Marathi, apply relationship-aware rules, and link to 1930 and Chakshu.

How is this different from Truecaller or Sanchar Saathi? Truecaller works on numbers; Sanchar Saathi's Chakshu is for reporting after the fact. We read the conversation itself, at the moment of decision, and we link to 1930 and Chakshu for reporting.

Other students have built scam classifiers. What is new? Most classify one SMS at a time. We track the conversation, know relationships (a delivery OTP from family versus a bank code), check claims against facts, and work in Marathi.

Technology

Why MuRIL and not ChatGPT or another LLM? It runs on the device, so chats stay private; it costs nothing per message; it answers in about 0.1 second; and it is trained for Indian languages. A cloud LLM would need to send private chats to a server.

Is this just keyword rules? No. The core is a fine-tuned neural model with four outputs. Rules handle what must never be missed (an OTP request), and the Brain reasons over the chat. The demo shows a case where every message scored low and the Brain still caught the scam.

Your data is synthetic. How do you know it works on real messages? I don't fully know yet, and I say so on the slides. We reduced the risk: an independent outside dataset, a test set used once, and a self-test written separately. The real-message test is the first pilot milestone.

Isn't 94% suspiciously high? Overfitting? The test messages were never seen in training, near-duplicates were kept on the same side of the split, and results agree with a separate self-test (97% and 3.4%). But because test data is synthetic, real-world numbers will likely be lower; that is why the pilot measures them.

What happens with a false alarm? The user sees a warning with reasons, never a block. They can tap "Not a scam", which quiets that chat until new strong evidence appears. We tuned for 3.3% false alarms and normal chats stay quiet below Dangerous.

What if it misses a scam? It is a safety layer, not a guarantee. Hard rules make sure OTP, PIN and "safe account" requests are never missed, and SOS is always available manually.

What if scammers change their wording? The normalizer defeats common tricks (look-alike letters, spacing). The model learns intent, not exact words. New patterns from pilot reports go into retraining, and the fact library is easy to update.

How does it handle Hinglish and Marathi typos? MuRIL was pre-trained on Roman-script Indian text, and our data includes typos, slang and emoji styles applied equally to scam and normal messages.

How do the explanations work? Are they faithful? Explanations come from what the model and Brain actually found: the intent pillars, rules, identifier checks and fact checks. They are written from verified templates in three languages, never generated freely.

How big is the model? Will it run on a phone? About 237 million parameters and 950 MB today. For phones we plan distillation and 8-bit quantisation, typically about 4× smaller, running on an Arm processor.

Privacy, law and deployment

How can it read WhatsApp chats? Is that allowed? WhatsApp doesn't let other apps read chats. The phone pilot uses two opt-in methods: the user shares a message to check, or allows reading of notifications. The full in-WhatsApp version needs a partnership with Meta or a government channel; that is the long-term goal.

What about privacy and the DPDP Act? Everything is analysed on the phone; nothing is uploaded. It follows consent, purpose limitation and data minimisation. Only messages a user chooses to share in the pilot, anonymised, are used to improve the model.

Is the scam registry real? In the demo it is simulated and labelled so. A real deployment needs access to I4C's Suspect Registry through Maharashtra Cyber.

What about iPhones? iOS restricts notification access, so the share-to-check mode is the iPhone route at first.

Impact, cost and plan

How will you measure impact? For each pilot stage: false alarms on real messages (target ⩽ 3%), scams caught (⩾ 90%), whether users understand alerts (⩾ 80%), and fewer users paying or sharing codes.

Who pays, and is it sustainable? Early stages: me and the college. Later: Maharashtra Cyber's awareness budget, bank CSR, or a work order. Running cost is about ₹0 per user because nothing runs in the cloud.

What would you deliver with a ₹10 lakh work order? A 3-month pilot with Pune City Cyber Police for about 5,000 users, an Android app with on-device AI, a consented real-message test set, and a public evaluation report.

Which department owns this? The Home Department through Maharashtra Cyber, starting with the Cyber Police Station, Pune City; MSInS for the technology-pilot route.

You and the process

You are alone. Can you deliver? The working prototype shows I can build it end to end. For the pilot I need mentorship and a partner office; volunteers run the community stages, as in the scale plan.

Did you use AI tools? Yes, and I say so on the slides. Claude helped write code and synthetic training messages. I designed the system, made and tested every decision through experiments, and ran all the training and evaluation.

What failed? Three things: our own data alone scored 0.75 on another writer's messages (fixed with outside data, 0.93); the model flagged a normal family fees message (fixed with 130 everyday money messages); and a learned conversation model lost to simpler rules, so we kept the rules.

What would you do differently? Start with real messages from users much earlier, and design the phone version from the start.

Weak spots and what not to say

Judges probe the weakest point; naming it first, with the plan to fix it, turns it into credibility.

Weak spot	Say this	Never say
No real-message data yet	"All tests so far are synthetic; the first pilot milestone is a consented real-message test."	"It works 94% in the real world."
Not inside WhatsApp yet	"Proposed integration; today it runs in our WhatsApp-style simulator; the phone pilot uses share-to-check and notifications."	"It is already integrated with WhatsApp."
Registry is simulated	"Simulated in the demo; a real one needs I4C data through Maharashtra Cyber."	"We use the government registry."
No stakeholder interviews	"Evidence today is official data; interviews and user testing start with the 20-person pilot."	Invented quotes or numbers
Two known misses	"A genuine festive offer was flagged, and a Hindi withdrawal-fee threat scored only Caution; both are in our next data round."	"It never makes mistakes."
Solo builder	"The prototype shows I can build end to end; for the pilot I need mentorship and a partner office."	"I can do it all alone."
AI assistance	"Claude helped with code and synthetic messages; I designed, tested and decided everything."	Hiding it
Model size	"950 MB on a laptop; compression for phones is planned."	"It already runs on phones."
Voice and video calls	"Out of scope for now: we cover WhatsApp text; digital-arrest video calls usually start with messages we can catch."	Claiming call detection

Words to avoid: "100% accurate", "guaranteed", "no privacy issues at all" (say "privacy by design: nothing leaves the phone"), "government-approved", "already deployed".

If you don't know an answer: "That's a good question; I haven't measured that yet. Here is how I would test it…" Then name one concrete test. Never guess a number.

Day-of timeline and rehearsal plan

Three full rehearsals before the slot are enough; sleep matters more than a fourth.

Tonight or this morning (about 2 hours):

- [ ] Read this doc once, fully; mark any answer you can't say in your own words
- [ ] Say the 60-second pitch aloud 5 times, until it flows without reading
- [ ] Rehearsal 1: full 5-minute pitch with the deck, timed
- [ ] Rehearsal 2: run all three demos with the real model; note anything that behaves differently
- [ ] Rehearsal 3: pitch + demo together, recorded on your phone; watch it once at 1.5× speed
- [ ] Ask someone at home to fire 5 questions from the Q&A section at you
- [ ] Optional: screen-record one clean demo run as your backup video

On the day:

When	Do
Morning	Check mail, portal and SMS again for the slot; reply in the thread if still nothing
Slot − 60 min	Eat, water, phone on silent except for calls from organisers
Slot − 30 min	Laptop on charger; close all apps; Focus mode on
Slot − 20 min	Start the app; confirm threshold 0.86; send the demo-1 warm-up, then reset
Slot − 15 min	Open the deck PDF and this doc; test mic, camera and screen share
Slot − 10 min	Join the meeting; name shown as `Anushka Bhakare - PRAHAARSHIELD`
During	Greet, ask how long you have, pitch, demo, then answers in three beats

When	Do
After	Write down every question you were asked; it helps for the next round

First words when the call starts: "Good morning. Thank you for this opportunity. How much time do I have for the presentation and the demo?" Then adjust: 3 minutes → short version; 5 or more → full version.

Body language: look at the camera, not the screen, when making your key points; smile at the start; pause one second before each answer; it is fine to say "let me show you" and go to the demo.

Image module: what we need and where images can ethically come from

The first image features need no model training, so they need a small, real test set, not a large training set.

Feature	How it works	Real images needed?
UPI QR check	Decode the QR with a standard library, read the payee and amount, check against the registry. A UPI QR always pays out; scanning never receives money	Only for testing; we can generate our own test QR codes
Screenshot and fake-notice reading	A pre-trained OCR engine (EasyOCR, which as far as I know supports Hindi, Marathi and English) extracts the text, which goes into the same text engine and Brain	About 50–100 real-looking screenshots to measure how well OCR reads Hindi and Marathi chat screenshots
An image classifier trained from scratch	Not planned: it would need thousands of real, consented, labelled images	Yes, many; a later phase with a pilot partner

Government-supplied images, your idea: good, and the right route is formal. Ask Maharashtra Cyber or I4C (through the pilot partnership) for an agreement to receive anonymised scam screenshots. Victim screenshots contain personal data, so names, numbers and account details must be removed, and sharing needs written permission under their data rules. Put it in the ask: "access to anonymised scam artefacts for a consented evaluation set". Do not expect it before the pilot.

Sources that really exist, and what each is good for:

Source	What it gives	Limits
Images you create yourself	Screenshots of your own test messages in different layouts, fonts and languages; your own UPI QR codes (payee and amount you choose), benign and malicious variants	Synthetic, so measures the pipeline, not real-world variety. Research datasets of QR codes were built the same way, from URLs with the `qrcode` library (Trad et al.)
Volunteers (family, friends, college)	Real scam screenshots they received, shared with consent, names and numbers blurred	Small number; needs a consent line
PhishTank and OpenPhish	Verified phishing website addresses; you capture the page screenshots yourself	Websites, not WhatsApp chats; check each record is still online and what the terms allow
Phish-IRIS dataset	1,313 training and 1,539 test screenshots of phishing web pages of 14 brands plus legitimate pages	Academic use only; web pages from 2018
CIC-Trap4Phish	Over one million generated QR images from malicious and benign URLs	Synthetic, URL-based; UPI payment QR codes have a different format

Source	What it gives	Limits
Official awareness material (I4C's CyberDost, RBI's awareness campaigns, Sanchar Saathi, Maharashtra Cyber and Mumbai Police posts)	Real examples of scam messages and notices, useful to understand patterns	They are public posts, not datasets: cite them and use only as pattern reference or with permission, never redistribute

What I have not found: a public dataset of real Indian WhatsApp scam screenshots. Say so if asked; it is why the government partnership is part of the plan.

Ethics line for the jury: no scraping of private chats; only consented, anonymised or self-generated images; official public material cited, not copied into a dataset; nothing uploaded to a server.

Will they ask to see the code?

The published process asks for the deck, plus an optional prototype or demo link; I found no rule that a code walkthrough is required. But your "Pilot / deployment details" field contains the GitHub link, so a juror may open it and ask how something is built. Be ready for a two-minute tour, and for "explain this file".

Tour order (about 2 minutes): open the repository README, then the folders.

Where	What it does	One sentence to say
`app/engine/normalizer.py`	Cleans tricks: look-alike letters, hidden characters, spaced letters	"First we undo the tricks scammers use to hide words."
`app/engine/entities.py`	Finds links, UPI IDs, phone numbers, amounts	"We replace them with placeholders so the model learns patterns, not specific numbers."
`app/engine/text_engine.py`	Loads MuRIL, runs the four heads, applies calibration, hard rules and the threshold	"This is the model: one encoder, four questions answered at once."

Where	What it does	One sentence to say
`app/engine/identifiers.py`	Checks links, UPI IDs and numbers against official lists and the registry	"This catches fake domains and numbers even when the text looks fine."
`app/brain/reasoning.py`	Tactics, chat memory, relationship rules, the 19 fact checks	"This is why it can explain and why it understands who is asking."
`app/brain/brain.py`	Turns everything into levels L0–L4, alerts and the SOS report	"This is the decision layer."
`app/server/main.py`, `app/frontend/sim.html`	FastAPI server and the WhatsApp-style demo	"The demo you just saw."
`training/`	Data preparation, the outside-data fetcher, style variants, the Colab notebooks	"Everything needed to retrain, including how we split data without leakage."

Questions about the code to be ready for:

- *Show me where the model is called.* `text_engine.py`: the function that tokenizes the cleaned text, runs the encoder, and returns scam probability, type, pillars and language.
- *What is the threshold and where is it set?* 0.86, stored in the model's config file, chosen on validation data only.
- *How do you stop data leakage?* `prepare_v3.py` groups near-duplicate messages so they stay in the same split, removes rows that resemble self-test cases, and strips hidden interface tags.
- *Which parts did the AI assistant write?* Be direct: the assistant helped write much of the code and synthetic messages; you designed the system, chose and ran the experiments, and made every decision. Make sure you can explain each file above in your own words; read each one once before the call.
- *Can I run it?* Yes: the README lists the steps; the model file is on a shared drive because it is about 1 GB.

Tonight, open each file in the table once, read the main function, and be able to say what goes in and what comes out.